

SidewalkPilot Steering Model Report

Combined current-label offline evaluation

Generated 2026-05-10 12:13:45. All available SidewalkPilot steering checkpoints are evaluated on the same active correction set: 1,464 labeled images from steering_corrections.json.

Active checkpoint naming is **SidewalkPilot-vX.Y.pth** for final checkpoints and **SidewalkPilot-vX.Yb.pth** for best checkpoints.

Best offline checkpoint: **2.3** (SidewalkPilot-v2.3.pth) with MAE 2.726, median AE 1.882, score 98.486%. Second: **2.3b** (SidewalkPilot-v2.3b.pth) with MAE 2.822.

Series 1 uses raw BGR and output scale 86. Series 2 uses output scale 85; v2.0/v2.0b use legacy HSV/CLAHE preprocessing, while v2.1 and newer use raw BGR.

Active Correction Sources

Source	Count	Purpose
D0328	315	First Dataset relabel, March 28
D0329	413	First Dataset relabel, March 29
D0425	65	Street test
D0426	53	Curves and shadows
D0427	72	Curved curb
D0429	53	Driveway and shadow
D0502_12	154	Shadow fix set
D0502_19	156	Hard turns, curb hugging, smoothness
D0503	159	Harsh sidewalk
D0506	24	8pm sidewalk failure set

Combined Model Ranking

Rank	Model	Checkpoint filename	Prep	MAE	Med	Max	Signed	<=5	<=10	Score
1	2.3	SidewalkPilot-v2.3.pth	raw BGR	2.73	1.88	25.11	0.04	1203/1464	1432/1464	98.486%
2	2.3b	SidewalkPilot-v2.3b.pth	raw BGR	2.82	1.96	24.83	0.50	1188/1464	1434/1464	98.432%
3	2.2b	SidewalkPilot-v2.2b.pth	raw BGR	4.20	3.24	41.55	0.08	1000/1464	1343/1464	97.665%
4	2.2	SidewalkPilot-v2.2.pth	raw BGR	4.22	3.30	35.45	0.25	984/1464	1350/1464	97.654%
5	2.0	SidewalkPilot-v2.0.pth	HSV/CLAHE -> BGR	4.93	4.93	149.97	-2.60	741/1464	938/1464	93.727%
6	2.0b	SidewalkPilot-v2.0b.pth	HSV/CLAHE -> BGR	4.99	4.99	149.49	-2.56	735/1464	932/1464	93.712%
7	2.1b	SidewalkPilot-v2.1b.pth	raw BGR	11.59	5.43	155.58	-0.99	657/1464	954/1464	93.564%
8	2.1	SidewalkPilot-v2.1.pth	raw BGR	11.66	5.39	156.37	-0.98	660/1464	953/1464	93.523%
9	1.9b	SidewalkPilot-v1.9b.pth	raw BGR	14.08	6.53	164.82	-2.31	668/1464	838/1464	92.178%
10	1.9	SidewalkPilot-v1.9.pth	raw BGR	14.14	6.65	164.79	-2.33	657/1464	832/1464	92.146%
11	1.8b	SidewalkPilot-v1.8b.pth	raw BGR	16.04	9.30	170.47	-3.16	580/1464	749/1464	91.090%
12	1.8	SidewalkPilot-v1.8.pth	raw BGR	16.04	9.37	170.49	-3.17	581/1464	749/1464	91.089%
13	1.7b	SidewalkPilot-v1.7b.pth	raw BGR	17.35	11.44	161.47	-3.03	497/1464	681/1464	90.361%
14	1.7	SidewalkPilot-v1.7.pth	raw BGR	17.44	11.53	160.93	-2.84	487/1464	680/1464	90.311%
15	1.5	SidewalkPilot-v1.5.pth	raw BGR	17.92	11.07	159.65	-3.47	459/1464	682/1464	90.046%
16	1.5b	SidewalkPilot-v1.5b.pth	raw BGR	17.92	11.07	159.65	-3.47	459/1464	682/1464	90.046%
17	1.6	SidewalkPilot-v1.6.pth	raw BGR	18.27	12.08	168.37	-2.96	429/1464	669/1464	89.852%
18	1.6b	SidewalkPilot-v1.6b.pth	raw BGR	18.29	12.00	167.80	-3.07	427/1464	670/1464	89.841%
19	1.4	SidewalkPilot-v1.4.pth	raw BGR	19.83	12.10	160.98	-4.70	414/1464	643/1464	88.985%
20	1.4b	SidewalkPilot-v1.4b.pth	raw BGR	20.15	12.65	160.94	-4.75	410/1464	645/1464	88.807%
21	1.2	SidewalkPilot-v1.2.pth	raw BGR	21.88	14.56	161.00	-5.10	339/1464	561/1464	87.844%
22	1.2b	SidewalkPilot-v1.2b.pth	raw BGR	21.91	14.62	161.00	-4.97	349/1464	565/1464	87.830%
23	1.3	SidewalkPilot-v1.3.pth	raw BGR	22.00	14.39	161.00	-5.33	331/1464	558/1464	87.778%
24	1.3b	SidewalkPilot-v1.3b.pth	raw BGR	22.24	14.27	161.35	-5.17	333/1464	573/1464	87.646%
25	1.1	SidewalkPilot-v1.1.pth	raw BGR	23.18	14.99	163.76	-6.13	307/1464	537/1464	87.125%
26	1.1b	SidewalkPilot-v1.1b.pth	raw BGR	23.18	14.99	163.76	-6.13	307/1464	537/1464	87.125%
27	1.0b	SidewalkPilot-v1.0b.pth	raw BGR	28.66	19.66	163.08	-5.66	213/1464	394/1464	84.075%
28	1.0	SidewalkPilot-v1.0.pth	raw BGR	29.21	20.76	161.51	-6.33	197/1464	376/1464	83.775%

Chronological Model Growth

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
1.0	SidewalkPilot-v1.0.pth	1	raw BGR	86	29.21	20.76	-6.33	80	197	707	89.97
1.0b	SidewalkPilot-v1.0b.pth	1	raw BGR	86	28.66	19.66	-5.66	85	213	741	90.65
1.1	SidewalkPilot-v1.1.pth	1	raw BGR	86	23.18	14.99	-6.13	144	307	879	90.17
1.1b	SidewalkPilot-v1.1b.pth	1	raw BGR	86	23.18	14.99	-6.13	144	307	879	90.17
1.2	SidewalkPilot-v1.2.pth	1	raw BGR	86	21.88	14.56	-5.10	171	339	909	91.20
1.2b	SidewalkPilot-v1.2b.pth	1	raw BGR	86	21.91	14.62	-4.97	168	349	925	91.33
1.3	SidewalkPilot-v1.3.pth	1	raw BGR	86	22.00	14.39	-5.33	158	331	912	90.97
1.3b	SidewalkPilot-v1.3b.pth	1	raw BGR	86	22.24	14.27	-5.17	153	333	901	91.13
1.4	SidewalkPilot-v1.4.pth	1	raw BGR	86	19.83	12.10	-4.70	211	414	971	91.60
1.4b	SidewalkPilot-v1.4b.pth	1	raw BGR	86	20.15	12.65	-4.75	223	410	953	91.55
1.5	SidewalkPilot-v1.5.pth	1	raw BGR	86	17.92	11.07	-3.47	287	459	995	92.84
1.5b	SidewalkPilot-v1.5b.pth	1	raw BGR	86	17.92	11.07	-3.47	287	459	995	92.84
1.6	SidewalkPilot-v1.6.pth	1	raw BGR	86	18.27	12.08	-2.96	240	429	995	93.34
1.6b	SidewalkPilot-v1.6b.pth	1	raw BGR	86	18.29	12.00	-3.07	242	427	999	93.23
1.7	SidewalkPilot-v1.7.pth	1	raw BGR	86	17.44	11.53	-2.84	303	487	1007	93.47
1.7b	SidewalkPilot-v1.7b.pth	1	raw BGR	86	17.35	11.44	-3.03	298	497	1019	93.27
1.8	SidewalkPilot-v1.8.pth	1	raw BGR	86	16.04	9.37	-3.17	395	581	1058	93.14
1.8b	SidewalkPilot-v1.8b.pth	1	raw BGR	86	16.04	9.30	-3.16	398	580	1054	93.14
1.9	SidewalkPilot-v1.9.pth	1	raw BGR	86	14.14	6.65	-2.33	441	657	1077	93.98
1.9b	SidewalkPilot-v1.9b.pth	1	raw BGR	86	14.08	6.53	-2.31	450	668	1082	93.99
2.0	SidewalkPilot-v2.0.pth	2	HSV/CLAE ->BGR	85	11.29	4.93	-2.60	464	741	1179	93.70
2.0b	SidewalkPilot-v2.0b.pth	2	HSV/CLAE ->BGR	85	11.32	4.99	-2.56	459	735	1179	93.74
2.1	SidewalkPilot-v2.1.pth	2	raw BGR	85	11.66	5.39	-0.98	325	660	1187	95.32
2.1b	SidewalkPilot-v2.1b.pth	2	raw BGR	85	11.59	5.43	-0.99	325	657	1188	95.31
2.2	SidewalkPilot-v2.2.pth	2	raw BGR	85	4.22	3.30	0.25	479	984	1454	96.55
2.2b	SidewalkPilot-v2.2b.pth	2	raw BGR	85	4.20	3.24	0.08	498	1000	1453	96.38
2.3	SidewalkPilot-v2.3.pth	2	raw BGR	85	2.73	1.88	0.04	757	1203	1461	96.35
2.3b	SidewalkPilot-v2.3b.pth	2	raw BGR	85	2.82	1.96	0.50	739	1188	1460	96.81

Field-Case / Subset MAE

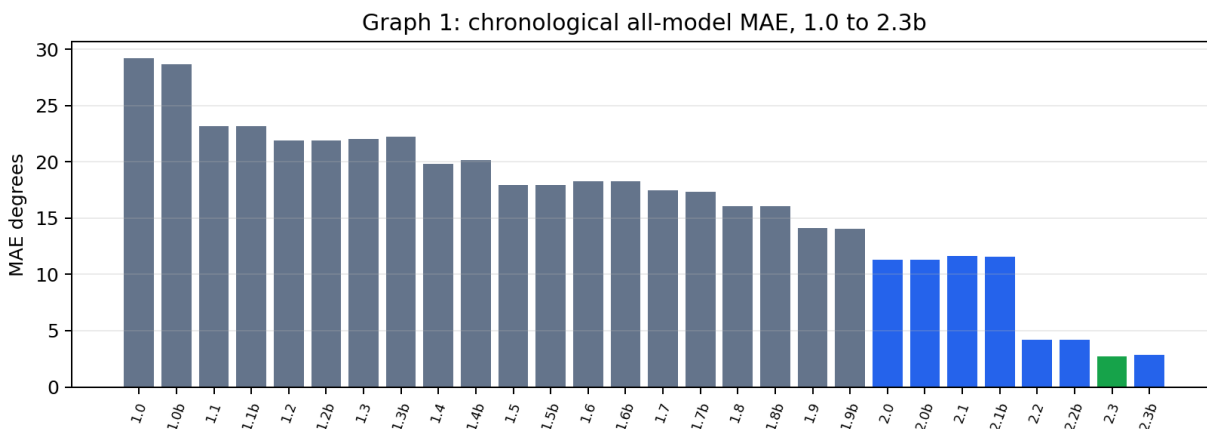
Model	Checkpoint filename	D0328	D0329	D0425	D0426	D0427	D0429	D0502_12	D0502_19	D0503	D0506
2.3	SidewalkPilot-v2.3.pth	2.39	2.51	3.30	3.04	3.46	2.54	3.23	3.07	2.38	3.71
2.3b	SidewalkPilot-v2.3b.pth	2.45	2.57	3.41	3.15	3.42	2.52	3.46	3.23	2.50	4.00
2.2b	SidewalkPilot-v2.2b.pth	3.81	3.56	5.46	4.97	5.61	4.38	5.67	4.72	3.50	2.62
2.2	SidewalkPilot-v2.2.pth	3.84	3.62	5.42	4.70	5.66	4.39	5.68	4.74	3.53	2.50
2.0	SidewalkPilot-v2.0.pth	17.97	20.51	2.71	2.03	2.57	1.69	2.35	2.26	1.66	36.00
2.0b	SidewalkPilot-v2.0b.pth	18.09	20.54	2.69	2.02	2.56	1.70	2.32	2.25	1.66	35.79
2.1b	SidewalkPilot-v2.1b.pth	19.65	19.96	3.65	3.19	3.81	3.33	4.10	3.64	2.42	3.64
2.1	SidewalkPilot-v2.1.pth	19.70	20.13	3.69	3.26	3.89	3.32	4.14	3.62	2.47	3.60
1.9b	SidewalkPilot-v1.9b.pth	17.10	21.79	1.66	1.49	1.73	1.43	1.36	1.78	28.24	35.89
1.9	SidewalkPilot-v1.9.pth	17.27	21.90	1.66	1.52	1.74	1.41	1.32	1.79	28.23	35.45
1.8b	SidewalkPilot-v1.8b.pth	17.43	20.92	1.52	1.19	1.42	0.97	1.08	22.58	28.02	37.04
1.8	SidewalkPilot-v1.8.pth	17.40	20.92	1.51	1.18	1.42	0.96	1.08	22.62	28.07	37.18
1.7b	SidewalkPilot-v1.7b.pth	16.52	20.73	1.39	1.03	1.22	1.01	17.43	22.68	27.59	30.87
1.7	SidewalkPilot-v1.7.pth	16.55	21.03	1.44	1.12	1.23	0.93	17.51	22.84	27.33	30.78
1.5	SidewalkPilot-v1.5.pth	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32
1.5b	SidewalkPilot-v1.5b.pth	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32
1.6	SidewalkPilot-v1.6.pth	16.84	20.09	1.26	0.84	1.10	26.32	18.91	22.95	27.36	29.10
1.6b	SidewalkPilot-v1.6b.pth	17.01	20.01	1.24	0.90	1.10	26.22	19.02	22.90	27.36	29.09
1.4	SidewalkPilot-v1.4.pth	16.34	20.23	0.88	0.56	31.68	25.83	18.73	24.90	26.43	34.05
1.4b	SidewalkPilot-v1.4b.pth	16.94	20.62	1.00	0.72	31.91	26.65	19.21	24.99	26.13	34.16
1.2	SidewalkPilot-v1.2.pth	16.38	19.88	0.87	39.03	31.76	25.38	20.86	23.31	32.51	36.89
1.2b	SidewalkPilot-v1.2b.pth	16.39	19.86	0.93	38.42	32.11	25.46	21.01	23.36	32.64	36.64
1.3	SidewalkPilot-v1.3.pth	16.46	20.12	0.87	37.56	34.06	25.26	19.51	23.50	34.04	32.99
1.3b	SidewalkPilot-v1.3b.pth	16.63	20.54	1.02	36.75	35.63	24.12	19.66	23.53	34.63	32.20
1.1	SidewalkPilot-v1.1.pth	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72
1.1b	SidewalkPilot-v1.1b.pth	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72
1.0b	SidewalkPilot-v1.0b.pth	17.36	20.30	49.19	45.39	54.06	37.39	29.16	30.30	40.52	40.61
1.0	SidewalkPilot-v1.0.pth	18.07	21.08	49.03	46.31	53.69	37.44	29.57	30.14	41.72	40.77

Prediction Distribution

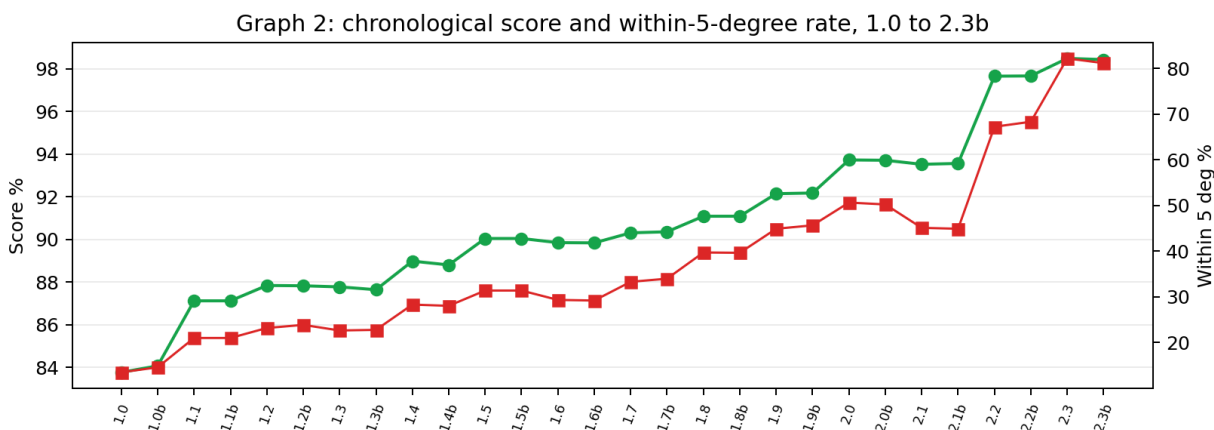
Model	Checkpoint filename	Pred min	P05	P25	Mean	Median	P75	P95	Pred max	Target mean
2.3	SidewalkPilot-v2.3.pth	5.00	45.72	80.45	96.35	91.49	103.33	174.80	175.00	96.30
2.3b	SidewalkPilot-v2.3b.pth	5.00	45.21	80.99	96.81	92.08	103.90	174.84	175.00	96.30
2.2b	SidewalkPilot-v2.2b.pth	5.00	42.88	79.96	96.38	91.77	105.08	174.68	175.00	96.30
2.2	SidewalkPilot-v2.2.pth	5.00	43.11	80.32	96.55	92.19	105.17	174.59	175.00	96.30
2.0	SidewalkPilot-v2.0.pth	5.00	36.16	72.59	93.70	90.20	111.83	171.22	175.00	96.30
2.0b	SidewalkPilot-v2.0b.pth	5.00	35.87	72.56	93.74	90.27	111.82	171.28	175.00	96.30
2.1b	SidewalkPilot-v2.1b.pth	5.00	40.19	75.04	95.31	91.84	111.06	172.32	175.00	96.30
2.1	SidewalkPilot-v2.1.pth	5.00	40.00	74.88	95.32	91.84	110.87	172.39	175.00	96.30
1.9b	SidewalkPilot-v1.9b.pth	4.00	39.32	74.31	93.99	89.93	111.05	168.06	176.00	96.30
1.9	SidewalkPilot-v1.9.pth	4.00	39.37	74.19	93.98	90.05	111.31	168.41	176.00	96.30
1.8b	SidewalkPilot-v1.8b.pth	4.00	40.87	74.96	93.14	90.60	107.88	160.84	176.00	96.30
1.8	SidewalkPilot-v1.8.pth	4.00	41.09	74.98	93.14	90.60	107.98	160.92	176.00	96.30
1.7b	SidewalkPilot-v1.7b.pth	4.00	40.95	75.29	93.27	91.09	108.71	158.73	176.00	96.30
1.7	SidewalkPilot-v1.7.pth	4.00	41.07	75.30	93.47	91.29	108.87	158.30	176.00	96.30
1.5	SidewalkPilot-v1.5.pth	4.00	41.51	75.46	92.84	91.06	107.40	158.62	176.00	96.30
1.5b	SidewalkPilot-v1.5b.pth	4.00	41.51	75.46	92.84	91.06	107.40	158.62	176.00	96.30
1.6	SidewalkPilot-v1.6.pth	4.00	42.91	74.73	93.34	91.77	107.75	160.57	176.00	96.30
1.6b	SidewalkPilot-v1.6b.pth	4.00	42.53	74.72	93.23	91.78	107.65	160.58	176.00	96.30
1.4	SidewalkPilot-v1.4.pth	4.00	41.55	75.76	91.60	90.91	104.66	150.01	176.00	96.30
1.4b	SidewalkPilot-v1.4b.pth	4.00	41.19	75.43	91.55	90.96	104.31	151.07	176.00	96.30
1.2	SidewalkPilot-v1.2.pth	4.00	42.51	72.93	91.20	91.20	107.34	146.78	176.00	96.30
1.2b	SidewalkPilot-v1.2b.pth	4.00	41.85	72.79	91.33	91.24	107.31	146.20	176.00	96.30
1.3	SidewalkPilot-v1.3.pth	4.00	41.83	74.14	90.97	90.35	105.54	145.66	176.00	96.30
1.3b	SidewalkPilot-v1.3b.pth	4.00	43.00	75.13	91.13	90.59	105.05	145.42	176.00	96.30
1.1	SidewalkPilot-v1.1.pth	4.00	39.48	73.08	90.17	89.60	104.79	150.05	176.00	96.30
1.1b	SidewalkPilot-v1.1b.pth	4.00	39.48	73.08	90.17	89.60	104.79	150.05	176.00	96.30
1.0b	SidewalkPilot-v1.0b.pth	9.01	39.89	71.87	90.65	90.44	108.41	144.09	173.93	96.30
1.0	SidewalkPilot-v1.0.pth	8.46	39.72	69.96	89.97	89.95	108.44	144.69	174.15	96.30

Prediction Buckets vs Ground

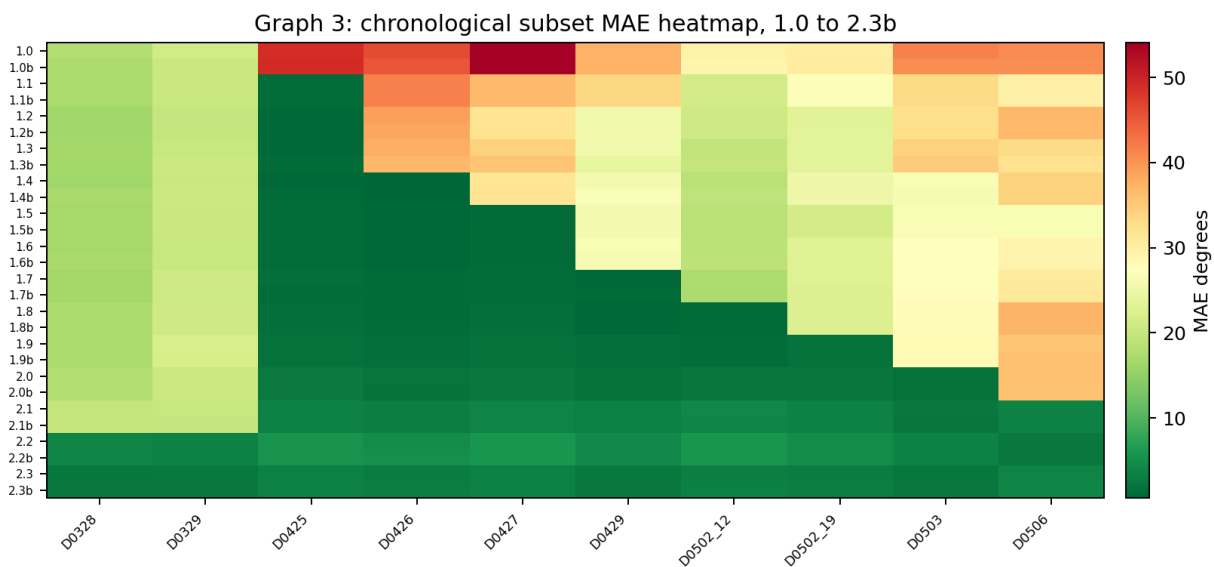
Model	Checkpoint filename	Pred HL 0-45	Pred L 45-75	Pred SL 75-85	Pred ST 85-95	Pred SR 95-105	Pred R 105-135	Pred HR 135-180	Bucket agree
2.3	SidewalkPilot-v2.3.pth	72	154	275	366	258	144	195	0/1464
2.3b	SidewalkPilot-v2.3b.pth	73	145	267	358	270	153	198	0/1464
2.2b	SidewalkPilot-v2.2b.pth	80	171	274	309	261	171	198	0/1464
2.2	SidewalkPilot-v2.2.pth	79	167	274	310	263	173	198	0/1464
2.0	SidewalkPilot-v2.0.pth	104	308	209	244	164	236	199	0/1464
2.0b	SidewalkPilot-v2.0b.pth	104	306	210	243	164	237	200	0/1464
2.1b	SidewalkPilot-v2.1b.pth	89	276	206	239	228	212	214	0/1464
2.1	SidewalkPilot-v2.1.pth	90	279	196	248	226	211	214	0/1464
1.9b	SidewalkPilot-v1.9b.pth	93	290	211	277	162	241	190	0/1464
1.9	SidewalkPilot-v1.9.pth	92	293	211	273	163	244	188	0/1464
1.8b	SidewalkPilot-v1.8b.pth	90	277	196	278	209	256	158	0/1464
1.8	SidewalkPilot-v1.8.pth	90	277	196	278	210	254	159	0/1464
1.7b	SidewalkPilot-v1.7b.pth	83	276	201	268	220	265	151	0/1464
1.7	SidewalkPilot-v1.7.pth	84	275	192	273	214	275	151	0/1464
1.5	SidewalkPilot-v1.5.pth	87	265	227	284	202	255	144	0/1464
1.5b	SidewalkPilot-v1.5b.pth	87	265	227	284	202	255	144	0/1464
1.6	SidewalkPilot-v1.6.pth	83	292	202	250	220	258	159	0/1464
1.6b	SidewalkPilot-v1.6b.pth	84	287	206	252	224	253	158	0/1464
1.4	SidewalkPilot-v1.4.pth	83	270	221	273	255	242	120	0/1464
1.4b	SidewalkPilot-v1.4b.pth	91	265	207	290	257	229	125	0/1464
1.2	SidewalkPilot-v1.2.pth	85	309	190	261	213	293	113	0/1464
1.2b	SidewalkPilot-v1.2b.pth	86	318	168	262	225	289	116	0/1464
1.3	SidewalkPilot-v1.3.pth	82	299	200	279	230	270	104	0/1464
1.3b	SidewalkPilot-v1.3b.pth	80	282	209	281	243	264	105	0/1464
1.1	SidewalkPilot-v1.1.pth	95	313	211	272	215	242	116	0/1464
1.1b	SidewalkPilot-v1.1b.pth	95	313	211	272	215	242	116	0/1464
1.0b	SidewalkPilot-v1.0b.pth	97	338	185	220	197	316	111	0/1464
1.0	SidewalkPilot-v1.0.pth	110	335	198	198	195	320	108	0/1464



Graph 1: chronological comparison using SidewalkPilot-vX.Y(b).pth checkpoints.



Graph 2: chronological comparison using SidewalkPilot-vX.Y(b).pth checkpoints.



Graph 3: chronological comparison using SidewalkPilot-vX.Y(b).pth checkpoints.

Notes

- All rows use the same current active correction set: 1,464 labeled images.
- Checkpoint filenames in this report use SidewalkPilot-vX.Y.pth for final checkpoints and SidewalkPilot-vX.Yb.pth for best checkpoints.
- Combined tables are decision comparisons; preprocessing differs for v2.0/v2.0b versus raw-BGR models.
- Offline MAE does not prove field reliability. Field tests still decide whether a model is safe enough to replace a rollback model.
- Graphs are chronological from 1.0 to 2.3b; ranking tables remain sorted by offline MAE.
- Lower MAE rows are colored greener and higher MAE rows trend toward yellow/red.